

Darukaa.Earth: AI Biodiversity Intelligence Chatbot Challenge

Welcome to the Darukaa.Earth Hackathon. This challenge is designed to evaluate your ability to build **intelligent, knowledge-driven AI systems** that can reason about real-world environmental problems.

We are not looking for UI-heavy applications. We are looking for **depth of thinking, system design, and scientific reasoning**.

Objective

Build an AI-powered conversational system that:

- Maintains a **structured knowledge base** of biodiversity and environmental metrics
- Understands user queries about ecosystems, land, and climate conditions
- Generates **actionable, non-obvious recommendations** to improve biodiversity
- Supports every recommendation with **scientific reasoning and evidence**

Core Requirements

1. Knowledge System (Critical)

Your system must include a **retrievable knowledge layer**, not just prompts.

It should cover:

- Soil health (pH, organic carbon, moisture)
- Land use / land cover
- Biodiversity indicators (species richness, habitat diversity)
- Climate factors (temperature, rainfall)
- Human impact (pollution, deforestation)

Expectation:

- Use RAG, embeddings, vector databases, or structured datasets
- Index research papers, reports, or environmental datasets
- Clearly show how knowledge is retrieved and used

2. Conversational Intelligence

Your system should:

- Ask **clarifying questions** when inputs are incomplete
- Handle **multi-turn conversations with memory**
- Adapt responses based on context

Example:

User: "Biodiversity is declining on my land"

System: "Can you provide soil organic carbon %, rainfall pattern, and land use type?"

3. Evidence-Backed Recommendations (Mandatory)

Each recommendation must include:

- **What to do**
- **Why it works (scientific reasoning)**
- **Which environmental metric improves**
- **Reference to a study/report/model**

Expected Quality:

"Introduce legume-based cover crops → increases soil organic carbon by ~15–25% over 2–3 years (FAO studies), improving microbial diversity and pollinator support."

Not acceptable:

"Use sustainable practices"

4. Multi-Metric Reasoning

Your system must connect multiple variables:

- Soil health ↔ biodiversity
- Water availability ↔ species survival
- Land use ↔ habitat fragmentation

This is the core differentiator—**no single-variable answers**.

5. Input Handling

Support at least:

- Text input (mandatory)
- Structured input (JSON or similar)

Bonus:

- Geo-coordinates or spatial context

6. Output Quality

Each response must clearly include:

- Recommendation
- Impacted metrics
- Time horizon (short / medium / long term)
- Confidence level (optional but valuable)

Example Use Case

Input:

- Soil organic carbon: 0.3%
- Rainfall: low
- Crop: monoculture wheat
- Region: semi-arid

Expected Output:

- Suggest agroforestry / intercropping
- Explain impact on soil carbon and biodiversity
- Provide measurable improvement estimates
- Reference credible sources such as
 - Food and Agriculture Organization
 - Intergovernmental Panel on Climate Change

Evaluation Criteria

1. Depth of Reasoning (30%)

- Are recommendations non-obvious?
- Do they combine multiple environmental variables?

2. Scientific Grounding (25%)

- Are claims backed by credible sources?
- Is reasoning accurate and explainable?

3. Knowledge System Design (20%)

- Use of RAG / vector DB / structured datasets
- Clarity of knowledge retrieval pipeline

4. Conversational Intelligence (15%)

- Context awareness
- Follow-up questioning
- Memory handling

5. Output Clarity (10%)

- Structured, readable, and actionable responses

Constraints

- No generic LLM-only solutions
- No shallow or obvious recommendations

- Must demonstrate **knowledge grounding + reasoning**
- Must handle **at least 3 environmental variables together**

Goal

Build a system that behaves like an **AI environmental scientist**, not a chatbot.

Contact

- **General Queries:** Ankita — +91 9330544618
- **Technical Queries:** Harsh — +91 93541 82834